

DS	Session	Searching	Question Information	Level 1	Challenge 3
Problem Description: Darsh recently found an new rectangular circuit board that he would like to recycle. That has R rows and C columns of squares. Each square of the circuit board has a very thinness, measured in millimetres. The square in the r-th row and c-th column has thinness $M_{r,c}$. A circuit board is beautiful if in each row, the difference between the thinnest square and the least thin square is no greater than L. Since the original circuit board might not be beautiful, Darsh would like to find a beautiful subcircuit board. This board can be obtained by choosing an axis-aligned subrectangle from the original board and taking the squares in that subrectangle. Darsh would like your help in finding the number of squares in the largest beautiful subrectangle of his original board. Constraints: $1 \leq T \leq 50$. $1 \leq R \leq 205$. $1 \leq C \leq 205$. $0 \leq M_i, j \leq 10^3$ for all i, j. $0 \leq L \leq 10^3$. Input Format: The first line of the input gives the number of test cases, T. T test cases follow. Each test case begins with one line containing three integers R, C and L, the number of rows, the number of columns, and the maximum difference in thinness allowed in each row. Then, there are R more lines containing C integers each. The c-th integer on the r-th line is M_r, c, the thinness of the square in the r-th row and c-th column. Output Format: Print the output in a separate lines contains the maximum number of squares in a beautiful subrectangle.					

Logical Test Cases

Test Case 1

INPUT (STDIN)

3
1 4 0
3 1 3 3
2 3 0
4 4 5
7 6 6
4 5 0
2 2 4 4 20
8 3 3 3 12
6 6 3 3 3
1 6 8 6 4

EXPECTED OUTPUT

2
2
6

Test Case 2

INPUT (STDIN)

3
2 4 0
2 3 1 2
3 1 3 3
2 3 0
41 4 5
7 61 6
4 5 0
12 12 14 4 20
81 13 13 3 12
61 16 31 3 3
11 16 81 6 4

EXPECTED OUTPUT

2
2
4

Mandatory Test Cases

Test Case 1

KEYWORD

int A[309][309];

Test Case 2

KEYWORD

bool ok[309][309][309];

Complexity Test Cases

Test Case 1

Test Case 2

Test Case 3

CYCLOMATIC COMPLEXITY

13

TOKEN COUNT

400

NLOC

53

```
Searching3.c +
1 #include <stdio.h>
2 #include <stdbool.h>
3
4 int A[309][309];
5 bool ok[309][309][309];
6
7 int main() {
8     int T;
9     if (scanf("%d", &T) != 1) return 0;
10    while (T--) {
11        int R, C, K;
12        scanf("%d %d %d", &R, &C, &K);
13        for (int i = 0; i < R; i++) {
14            for (int j = 0; j < C; j++) {
15                scanf("%d", &A[i][j]);
16            }
17        }
18        for (int i = 0; i < R; i++) {
19            for (int c1 = 0; c1 < C; c1++) {
20                int mn = A[i][c1];
21                int mx = A[i][c1];
22                for (int c2 = c1; c2 < C; c2++) {
23                    if (A[i][c2] < mn) mn = A[i][c2];
24                    if (A[i][c2] > mx) mx = A[i][c2];
25                    if (mx - mn <= K) {
```

```

26         ok[i][c1][c2] = true;
27     } else {
28         ok[i][c1][c2] = false,
29     }
30 }
31 }
32 }
33 int max_area = 0;
34 for (int c1 = 0; c1 < C; c1++) {
35     for (int c2 = c1; c2 < C; c2++) {

```

main | CH.SC.U4CSE25225 | Searching3.c

Ln 53, Col 1 Spaces: 4 UTF-8 CRLF C

```
36     int width = c2 - c1 + 1;
37     int current_h = 0;
38     for (int r = 0; r < R; r++) {
39         if (ok[r][c1][c2]) {
40             current_h++;
41             int area = current_h * width;
42             if (area > max_area) max_area = area;
43         } else {
44             current_h = 0;
```

```
45         }
46     }
47 }
48 }
49     printf("%d\n", max_area);
50 }
51 return 0;
52}
53
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

3

1 4 0

3 1 3 3

2 3 0

4 4 5

7 6 6

4 5 0

2 2 4 4 20

8 3 3 3 12

6 6 3 3 3

1 6 8 6 4

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching3.exe

2

2

6

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

3

2 4 0

2 3 1 2

3 1 3 3

2 3 0

41 4 5

7 61 6

4 5 0

12 12 14 4 20

81 13 13 3 12

61 16 31 3 3

11 16 81 6 4

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching3.exe

2

2

4

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>